Department of Mathematics, IIT Madras

MA 2102 - Differential Equations, Coordinator: Prof. A. K. B. Chand

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Assignment - 8 : First order PDEs and Canonical Form of 2nd order PDEs

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1. Form the partial differential equations by eliminating the arbitrary constants or arbitrary functions from the following relations:

   (a) $u = (x^2 + a)(y^2 + b)$;
   (b) $u = ax + by + ab$;
   (c) $(x-a)^2 + (y-b)^2 + u^2 = 1$;
   (d) $u = (x-a)^2 + (y-b)^2$;
   (e) $u = x^n f(y/x)$;
   (f) $u = e^{mx} f(x+y)$;
   (g) $G(xy + u^2,\ x + y + u) = 0$;
   (h) $xyu = f(x+y+u)$;
   (i) $G(x+y+u,\ x^2 + y^2 - u^2) = 0$;
   (j) $u = f(x+iy) + g(x-iy)$.

2. Solve each of the following using Characteristics / Lagrange Multiplier Method, where $p = u_x$ and $q = u_y$:

   (a) $x^2 p + y^2 q = (x+y)u$;
   (b) $u(x+y)p + u(x-y)q = x^2 + y^2$;
   (c) $x^2 p + q = u^2$;
   (d) $(x+1)^2 p + (y-1)^2 q = (x+y)u$;
   (e) $\cos(x+y)p + \sin(x+y)q = u$;
   (f) $(x^2 + y^2)p + 2xyq = (x+y)u$;
   (g) $xup + yup = xy$;
   (h) $y^2 p - xyq = x(u - 2y)$;
   (i) $p + 3q = 5u + \tan(y - 3x)$;
   (j) $u(u^2 + xy)(px - qy) = x^4$;
   (k) $py + qx = xyu^2(x^2 - y^2)$;
   (l) $(x^2 - y^2 - u^2)p + 2xyq = 2xu$;
   (m) $xyp + y^2 q = xyu - 2x^2$;
   (n) $u(p-q) = u^2 + (x+y)^2$;
   (o) $x(y^2 - u^2)p - y(u^2 + x^2)q = u(x^2 + y^2)$;
   (p) $(y+u)p + (u+x)q = x + y$;
   (q) $(y - ux)p + (x + yu)q = x^2 + y^2$;
   (r) $(x-y)p + (y - x - u)q = u$;
   (s) $(y-u)p + (u-x)q = x - y$;
   (t) $x(y-u)p + y(u-x)q = u(x-y)$;
   (u) $(x^2 - yu)p + (y^2 - ux)q = u^2 - xy$;
   (v) $p\tan x + q\tan y = \tan u$;
   (w) $(x + 2u)p + (4ux - y)q = 2x^2 + y$;
   (x) $(y^2 + u^2)p - xyq + xu = 0$;
   (y) $(x + y - u)p + (y + u - x)q = u + x - y$;
   (z) $(y+u)p + (u-x)q = u + 5y$.

3. Solve the following initial value problems using Lagrange's method:

   (a) $u_x + u_y - u = e^x, \quad u(x,0) = 0$;
   (b) $xu_x + yu_y = xe^{-u}, \quad u(x,x^2) = 0$;
   (c) $u_x - u_y + u = y, \quad u(x,0) = x^2$;
   (d) $xu_x - yu_y + u = x, \quad u(x,x) = x^3$;
   (e) $x(y^2 + u)u_x - y(x^2 + u)u_y = (x^2 - y^2)u, \quad u(x,-x) = 1$;
   (f) $u_x - u_y + \sqrt{2}\,u = 0, \quad u(x,0) = x$;
   (g) $u_x - u_y - \sqrt{2}\,u = 0, \quad u(x,0) = x^2$;
   (h) $u_x - u_y + \sqrt{2}\,u = 0, \quad u(x,0) = x + e^x$;
   (i) $u_x - u_y - \sqrt{2}\,u = 0, \quad u(0,y) = e^y - y$;
   (j) $3u_x - 4u_y + 2u = 0, \quad u(x,0) = e^x$;
   (k) $uu_x + u_y = 0, \quad u(x,0) = x$;

(l) $u_x + u_y = u^2, \quad u(x,0) = \cos x;$

(m) $xu_x + yu_y = u, \quad u(1,y) = 1 + y;$

(n) $u_x + 2xu_y = y, \quad u(0,y) = y^2;$

(o) $yu_x - xu_y = 0, \quad u(0,y) = y^2;$

(p) $u(x+y)u_x + u(x-y)u_y = x^2 + y^2, \quad u = 0$ on $y = 2x;$

(q) $u_x + u_y = 1$ with $u = x$ on $y = x$. Show that this problem has infinitely many solutions, and that with $u = 0$ on $y = x$ it has none. Explain the reason geometrically.

4. Discuss the type (hyperbolic, parabolic, or elliptic) of the following second-order linear equations:

(a) $u_{xx} + (x-1)u_{xy} + u_{yy} - 2x^2u_x + 3xyu_y + 2u = \sin x;$

(b) $yu_{xx} - xu_{xy} + yu_{yy} - 3x^2u_y = x\exp(-xy);$

(c) $(x+2)u_{xx} + 2(x+y)u_{xy} + 2(y-1)u_{yy} - 3x^2u_x = x^3y^3;$

(d) $y^2u_{xx} - x^2u_{yy} = 0, \quad x > 0,\ y > 0;$

(e) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = 0;$

(f) $u_{xx} + x^2u_{yy} = 0;$

(g) $u_{xx} + 2u_{xy} + 5u_{yy} + u_x = 0;$

(h) $u_{xx} + 2xu_{xy} + (1-y^2)u_{yy} = 0;$

(i) $yu_{xx} + u_{yy} = 0$ (Tricomi equation);

(j) $(1+x^2)u_{xx} + (1+y^2)u_{yy} + xu_x + yu_y = 0;$

(k) $\sin^2 x\, u_{xx} + \sin 2x\, u_{xy} + \cos^2 x\, u_{yy} = x;$

(l) $xu_{xx} + yu_{yy} + 3y^2u_x = 0;$

(m) $u_{xx} + yu_{xy} + xu_{yy} = 0;$

(n) $yu_{xx} + 2u_{xy} + u_{yy} = uu_x + u_y^2.$

In each case sketch the regions of the xy-plane where the equation is hyperbolic, parabolic and elliptic.

5. Reduce the following equations to their canonical forms and find their general solutions if possible.

(a) $u_{xx} + 2u_{xy} + u_{yy} = 0;$

(b) $u_{xx} + yu_{yy} = 0, \quad y < 0;$

(c) $u_{xx} - x^2u_{yy} = 0;$

(d) $u_{xx} + xu_{yy} = 0, \quad x < 0;$

(e) $u_{xx} - 2xyu_{xy} + x^2y^2u_{yy} = 0;$

(f) $x(xy-1)u_{xx} - (x^2y^2-1)u_{xy} + y(xy-1)u_{yy} + (x-1)u_x + (y-1)u_y = 0;$

(g) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y = 0;$

(h) $y^2u_{xx} - 2xyu_{xy} + x^2u_{yy} - \frac{y^2}{x}u_x - \frac{x^2}{y}u_y = 0;$

(i) $(y-1)u_{xx} - (y^2-1)u_{xy} + y(y-1)u_{yy} + u_x - u_y = 2ye^{2x}(1-y)^3;$

(j) $x^2(y-1)u_{xx} - x(y^2-1)u_{xy} + y(y-1)u_{yy} + xyu_x - u_y = 0;$

(k) $4u_{xx} - 7u_{xy} - 2u_{yy} = 243(2x+y)(x-4y)^2;$

(l) $3u_{xx} + u_{xy} - 2u_{yy} - 30u_x + 20u_y = 0;$

(m) $u_{xx} + 8u_{xy} + 16u_{yy} = 16;$

(n) $16u_{xx} - 24u_{xy} + 9u_{yy} + 36u_x - 27u_y = 0;$

(o) $3u_{xx} + 10u_{xy} + 3u_{yy} = 0;$

(p) $u_{xx}+4u_{xy}+3u_{yy}=0$;

(q) $u_{xx}+u_{xy}+u_{yy}=0$;

(r) $u_{xx}+2u_{xy}+17u_{yy}=0$;

(s) $u_{xx}+2u_{xy}+u_{yy}=4xy$;

(t) $u_{xx}-y^4u_{yy}=2y^3u_y, \quad y\neq 0$;

(u) $u_{xx}-xu_{yy}=\dfrac{u_x}{2x}, \quad x>0$;

(v) $y^2u_{xx}+u_{yy}=0, \quad y\neq 0$;

(w) $x^2u_{xx}+y^2u_{yy}=0, \quad x\neq 0,\ y\neq 0$;

(x) $4y^2u_{xx}-e^{2x}u_{yy}=0, \quad y\neq 0$;

(y) $y^2u_{xx}-x^2u_{yy}=0, \quad x>0,\ y>0$;

(z) $u_{xx}-2\sin x\,u_{xy}-\cos^2 x\,u_{yy}-\cos x\,u_y=0$;

6. Miscellaneous problems:

(a) Find the complete integral and the singular integral of $u=px+qy+pq$.

(b) Show that the integral surfaces of $yp-xq=0$ are surfaces of revolution about the u-axis. Find the integral surface containing the curve $u=y^2$, $x=0$.

(c) Find an integral surface of $xu_y-yu_x=0$ containing the circle $u=1$, $x^2+y^2=4$. Is it unique? Justify your answer.

(d) Show that if two integral surfaces of $a(x,y,u)p+b(x,y,u)q=c(x,y,u)$ intersect along a curve, then the curve is a characteristic curve.

(e) Under the change of variables $\xi=\xi(x,y)$, $\eta=\eta(x,y)$ with Jacobian $J\neq 0$, show that the discriminants satisfy $B^2-4AC=J^2(b^2-4ac)$. Deduce that the type of a second order linear PDE is invariant.

(f) Show that the general solution of $u_{xx}-2u_{xy}+u_{yy}=0$ is $u=\phi(x+y)+x\,\psi(x+y)$, where ϕ and ψ are arbitrary functions.

(g) Reduce $u_{\xi\eta}-4u_\xi+u_\eta+3u=0$ to a form free of first order derivatives by the substitution $u=v\,e^{\lambda\xi+\mu\eta}$.

(h) Reduce the Tricomi equation $yu_{xx}+u_{yy}=0$ to canonical form separately in the regions $y>0$ and $y<0$.

(i) Reduce $u_{xx}+u_{yy}+2u_x-4u_y+u=0$ to the form $v_{xx}+v_{yy}+cv=0$, and find the constant c.